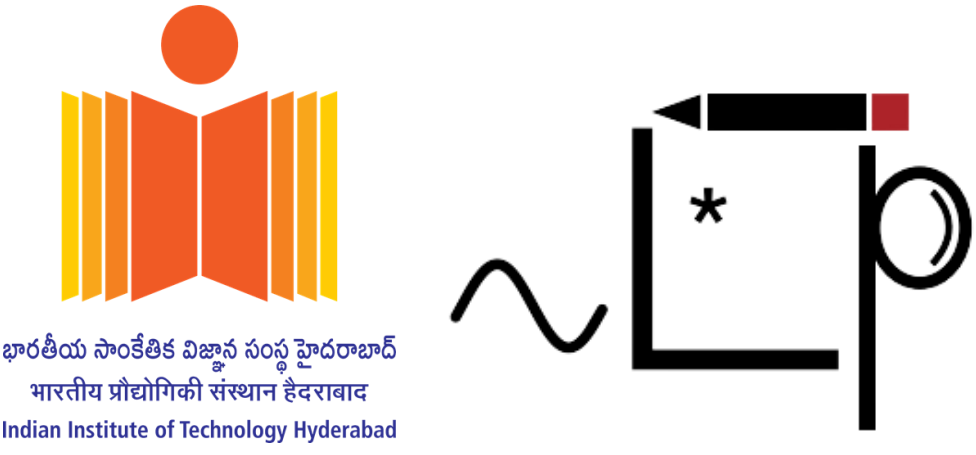


Aerial Mirage: Unmasking Hallucinations in Large Vision Language Models



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Motivation

- Integrating the capabilities of Large Vision Language Models (LVLMs) with drone images can improve description utility, benefiting applications like surveillance, drone-based delivery services, etc.
- How effectively can LVLMs generate relevant information from drone imagery?
- Can we use GPT4 and Gemini as judges for hallucination evaluation?

Challenges

- Traditional image captioning models **struggle with aerial imagery** due to diverse orientations, angular perspectives, small and unclear objects.
- Lack of image-caption datasets for drone imagery** poses a significant challenge for training and evaluating drone image captioning.
- Hallucination** in LVLMs impedes them from being used for practical applications.
- Hallucination Detection** in LVLMs.

Contributions

- First **Aerial-view Image Captioning dataset (AeroCaps)**, containing four captions per image.
- First **extensive analysis of hallucinations on aerial imagery** using different prompts and LVLMs on our proposed dataset and VisDrone. We explore the reasons behind such hallucinations.
- Release the generated captions from the LVLMs and **the annotated hallucination details** as the **Labelled Illusion Dataset (LID)**.
- Evaluation of the degree of hallucination in descriptions generated by LLaVA by models **GPT-3.5-turbo** (text), **GPT-4** (text-only), **Gemini-pro** (text) and **Gemini-pro-vision** (text + image) to check the reliability of these models as a judge.

Hallucination Detection Process

- Our initial question: **“How effectively can these LVLMs generate relevant information from drone imagery?”**
- Models: **LLaVA** and **InstructBLIP**
- Prompt: **“Describe the image briefly”** and **“Describe the image in less than 30 words”**.
- Dataset: **AeroCaps** and **VisDrone** (validation set)
- This results in multiple (Dataset, Model, Caption-length-type) views of the data.

AeroCaps Dataset



- A road with lots of trees and buildings in the background.
- A road with green trees passing through the tall buildings.
- An aerial view of the city with road passing through the tall buildings.
- A road is passing through the lush green field and tall buildings.

Category	Count
Building	267
Roundabouts	30
Garden	93
Vehicles	72
Parking space	68
Events	86
Agricultural field	78
Construction site	249
Road	182
Sports/Playground	37
Bus stop	10
Activity/ people	60
Campus Overview	24

Table 1: Different categories of the images in AeroCaps

Hallucination Categories & Labelled Illusion Dataset

- Any Hallucination, Object and Position Hallucination, Hallucinations due to Count error.
- Hallucination Levels (Minor, Moderate and Extreme), Names and number of Hallucinated Categories, Names and number of Missing Objects, Position of hallucinatory words.

An example of Object Hallucination



A **red car** driving down a road next to a bush.

An example of Position and Count Hallucination



The image depicts a man riding a motorcycle down a street, passing by a group of people standing on the sidewalk. There are a **total of five people** visible in the scene, with some standing closer to the motorcycle rider and others further away. **In addition to the motorcycle, there are two other motorcycles parked on the side of the street, one closer to the left side and the other further back...**

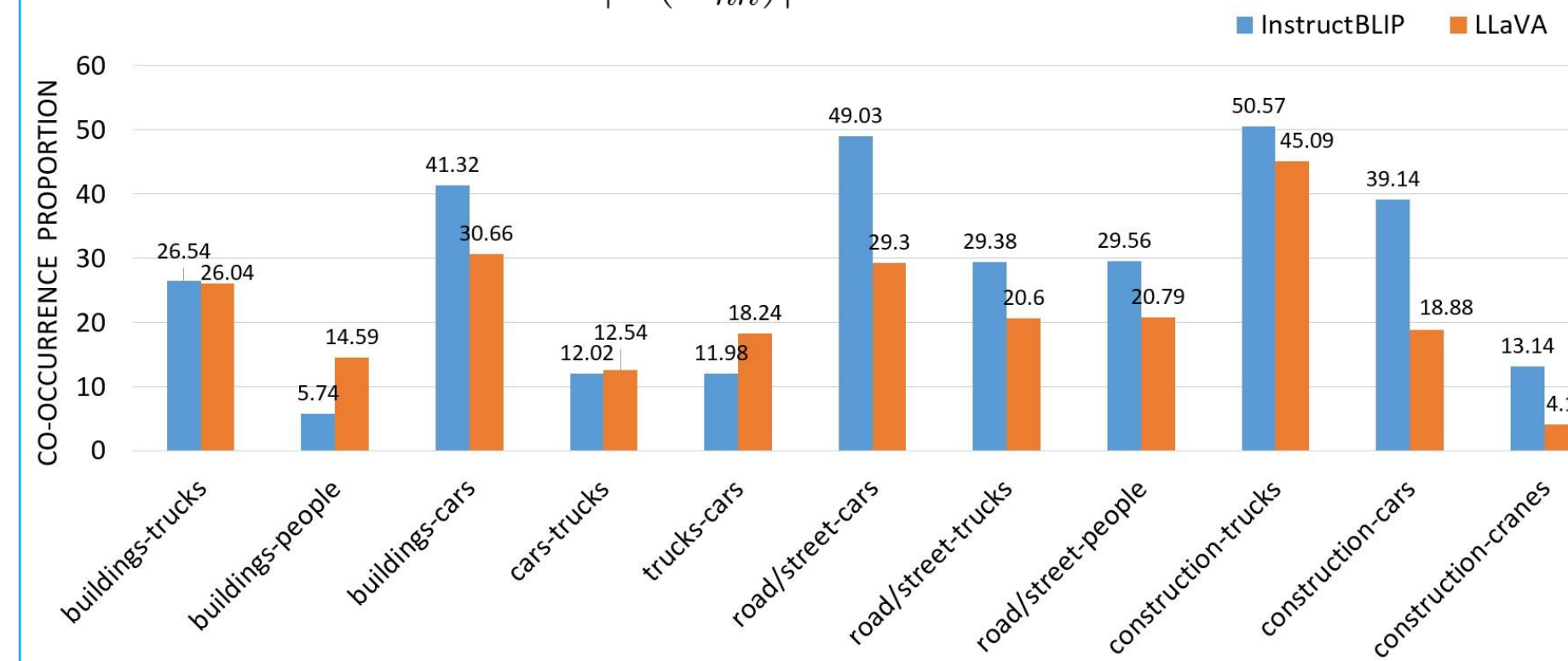
Results & Findings

Dataset	Model	Caption length type choice	CH_s	CH_i	Len	Pr	Re	F1
AeroCaps	LLaVA	Long description	71.38	12.84	100.7	92.14	76.8	82.94
AeroCaps	LLaVA	Short description	26.11	33.20	12.7	95.1	47.56	62.08
AeroCaps	InstructBLIP	Long description	74.36	13.89	106.3	89.79	71.71	78.97
AeroCaps	InstructBLIP	Short description	24.36	42.25	10	95.82	41.13	56.51
VisDrone	LLaVA	Long description without any additional information	63.5	9.74	108.8	94.45	77.74	85.03
VisDrone	LLaVA	Long description with hints about objects present	66.6	9.25	94.9	94.19	82.94	88.05
VisDrone	InstructBLIP	Long description	66.24	10.98	107.8	93.33	78.32	84.71

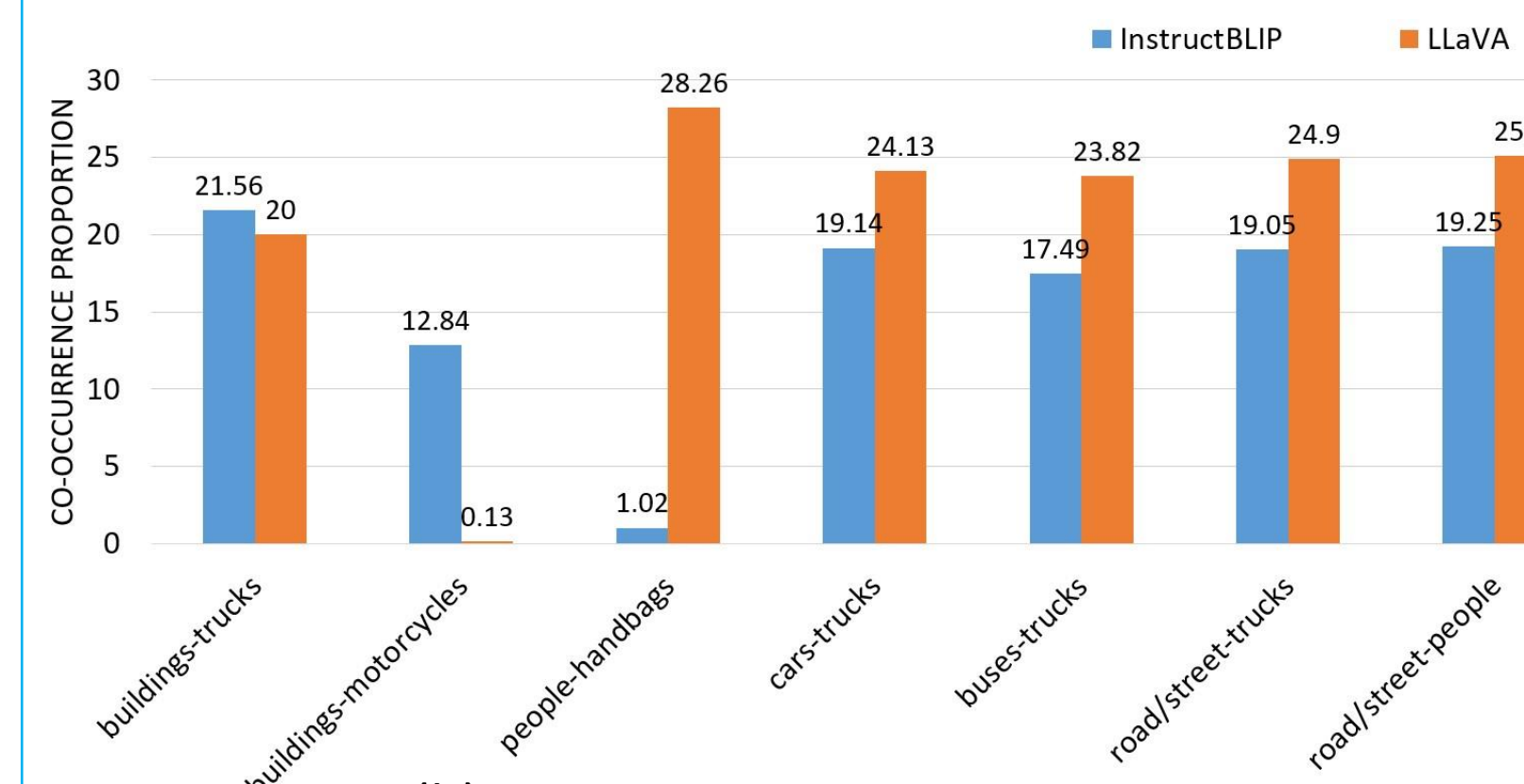
Table 2: CHAIR scores with different setups. CH_s and CH_i represents CHAIR_s and CHAIR_i, respectively. *Len* denotes the average length of the captions. *Pr*, *Re* and *F1* are Precision, Recall and F1 Score, respectively.

Co-Occurrence Proportion (CP)

$$CP = \frac{|\mathcal{A}(O_{nh}) \cap \mathcal{A}(O_h)|}{|\mathcal{A}(O_{nh})|} \times 100$$



(a) For AeroCaps Dataset



(b) For VisDrone Dataset

Fig: Co-occurring hallucinations among object pairs.

Position Index Ratio (PIR)

$$PIR_i = \frac{Index(W_i)}{N}$$

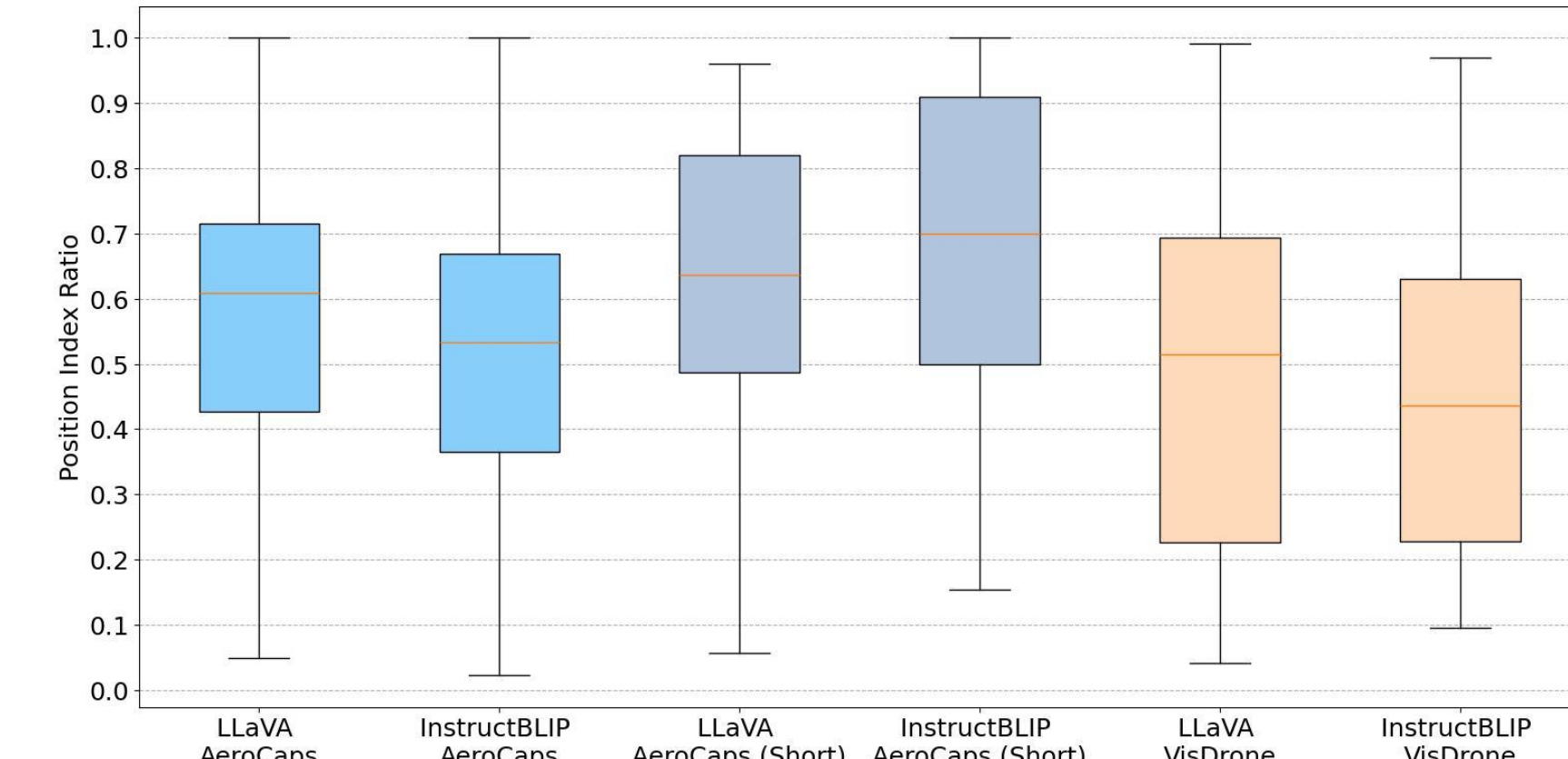


Fig: Position Index Ratios of all hallucinatory words. The yellow line denotes the median value.

Can we use GPT4 and Gemini as judges for hallucination evaluation?

Model	Correct%	Incorrect%
GPT-3.5-Turbo (text)	51.74	48.26
GPT-4 (text)	72.12	27.88
Gemini-pro (text)	41.29	58.71
Gemini-pro-vision (text + img)	20.40	79.60

Table 3: Hallucination judgement by advanced models